

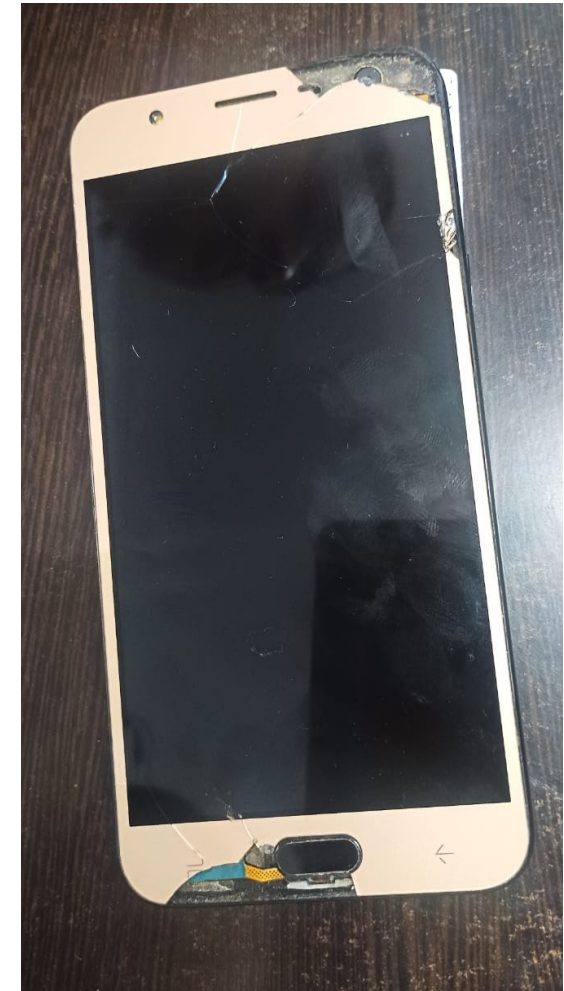
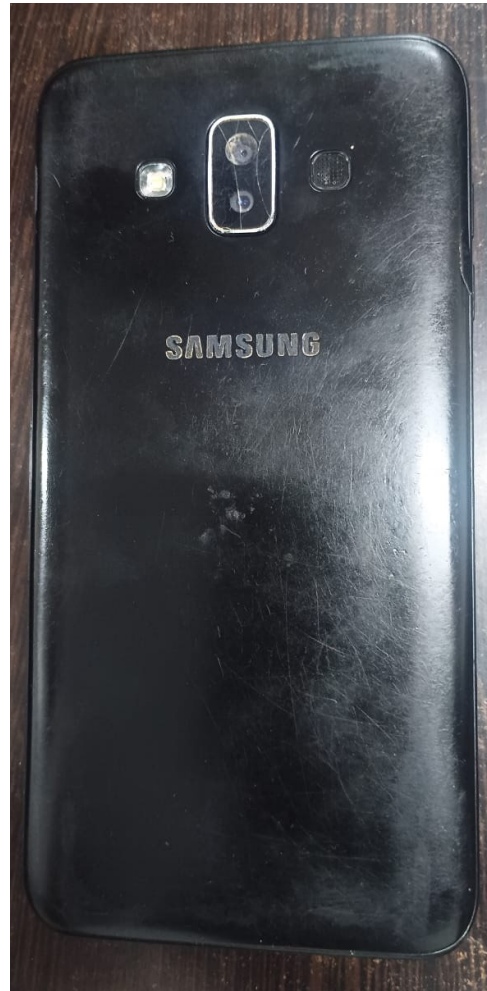
REVERSE ENGINEERING : SMARTPHONE

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EXTERNAL PRODUCT INSPECTION

Samsung Galaxy J7 Duo

- Manufacturer: Samsung
- Smartphone with 5.5" display
- Physical buttons and touchscreen
- Rear dual-camera system
- Micro-USB
- Removable rear cover and battery



PRODUCT LITERATURE & VERSION REVIEW

Parameter	Specification
Processor	Exynos 7885
CPU	Octa-core
GPU	Mali-G71
RAM	4 GB
Storage	32 GB
Expandable storage	microSD up to 256 GB
Display	5.5" HD AMOLED
Battery	3000 mAh
OS at launch	Android 8.0 Oreo
Connectivity	2G / 3G / 4G LTE

PURPOSE, USER NEEDS & TRADE-OFFS

Problem Solved

- Combines **communication, internet, photography and computing** in one portable device.
- Supports calls, messaging, applications and multimedia.

User Expectations

- Good battery life
- Reliable performance
- Good display
- Good camera
- Sufficient storage
- Reliable connectivity
- Easy operation

Engineering Trade-offs

- **Performance ↔ Battery consumption**
- **Large display ↔ Portability**
- **Features ↔ Cost**
- **Battery capacity ↔ Weight/thickness**
- **Camera system ↔ Internal space**

These trade-offs follow the reverse-engineering framework in your PDF; the specific Galaxy J7 Duo examples are engineering inferences.

TEARDOWN PROCESS & SAFETY

Tools

- Precision screwdriver set
- Plastic pry tools
- Spudger
- Tweezers
- ESD wrist strap
- ESD mat
- Magnifying glass
- Camera/phone
- Multimeter
- Screw containers
- Labels/markers

Safety

- Power off the phone
- Remove the battery before PCB work
- Use ESD protection
- Hold PCBs by their edges
- Avoid touching IC pins
- Keep screws organized



MECHANICAL DISASSEMBLY

External Mechanical Parts

- Removable rear cover
- Removable 3000 mAh battery
- Front display assembly
- Physical Home button
- Power button
- Volume buttons
- SIM card slots
- microSD slot
- Micro-USB port
- 3.5 mm audio jack
- Speaker openings
- Camera openings

Internal Mechanical Features

- PCB mounting points
- Screws
- Plastic clips
- Flexible cables
- Battery contacts
- Camera modules
- Antenna structures
- Display connectors



COMPONENT INVENTORY

Component

Main PCB
Exynos 7885 SoC
RAM
Flash storage
Battery
Display
Rear cameras
Front camera
Fingerprint sensor
Accelerometer
Speaker
Microphone
USB connector
Antennas
Power-management ICs

Function

Contains electronic circuits
Main processing
Temporary data storage
Permanent data storage
Power source
Visual output
Image capture
Selfies/video
Biometric security
Motion detection
Audio output
Audio input
Charging/data
Wireless communication
Voltage/power regulation



COMPONENT IDENTIFICATION & SPECIFICATIONS

Main Processor

Exynos 7885

- Manufacturer: Samsung
- Process technology: 14 nm
- CPU: Octa-core
- GPU: Mali-G71
- Used as the main SoC

Memory

- RAM: **4 GB**
- Internal storage: **32 GB**
- Expandable storage: **microSD up to 256 GB**

Display

- Type: Super AMOLED/HD AMOLED
- Size: 5.5 inches
- Resolution: 1280 × 720

COMPONENT FUNCTIONAL SPECIFICATIONS

Processor

Function: Executes instructions and controls system operation.

RAM

Function: Temporarily stores data used by running applications.

Flash Memory

Function: Stores Android, applications and user files.

Battery

Specification: 3000 mAh Li-ion removable

Function: Provides electrical power.

Cameras

Rear: 13 MP + 5 MP depth camera

Front: 8 MP

Function: Image/video capture.

Sensors

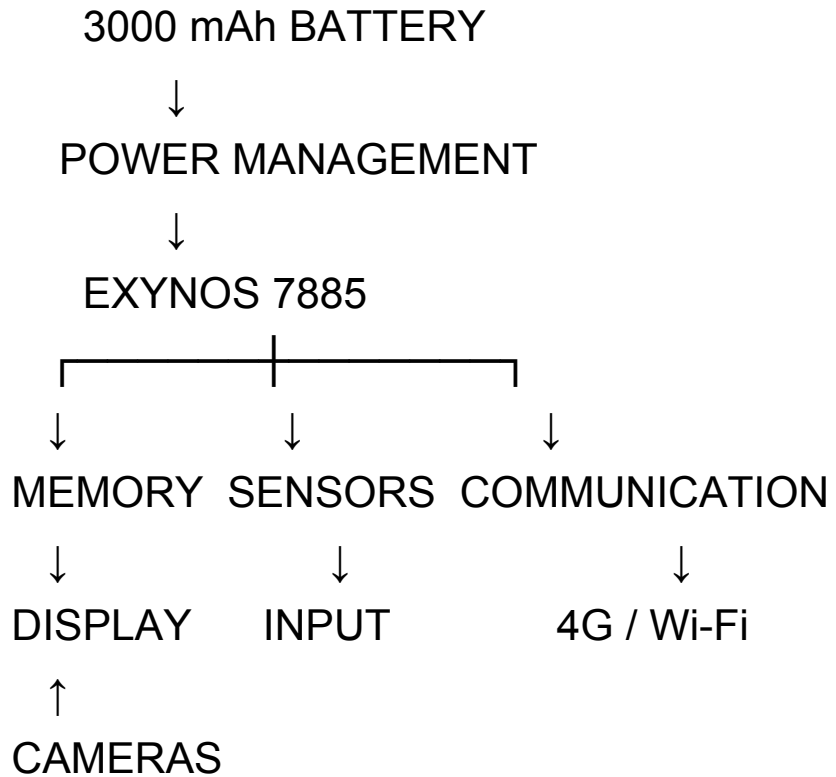
- Accelerometer
- Proximity sensor
- Fingerprint sensor

Display

Function: Provides visual information and touchscreen interaction.

Samsung's published FAQ confirms the battery, cameras and sensors.

SYSTEM ARCHITECTURE



INPUT–PROCESS–OUTPUT ANALYSIS

Inputs

- Touchscreen
- Physical buttons
- Camera
- Microphone
- Fingerprint sensor
- Light sensor
- Cellular signals
- Wi-Fi signals
- Bluetooth signals



Processing

- Exynos processor + RAM + operating system + supporting ICs

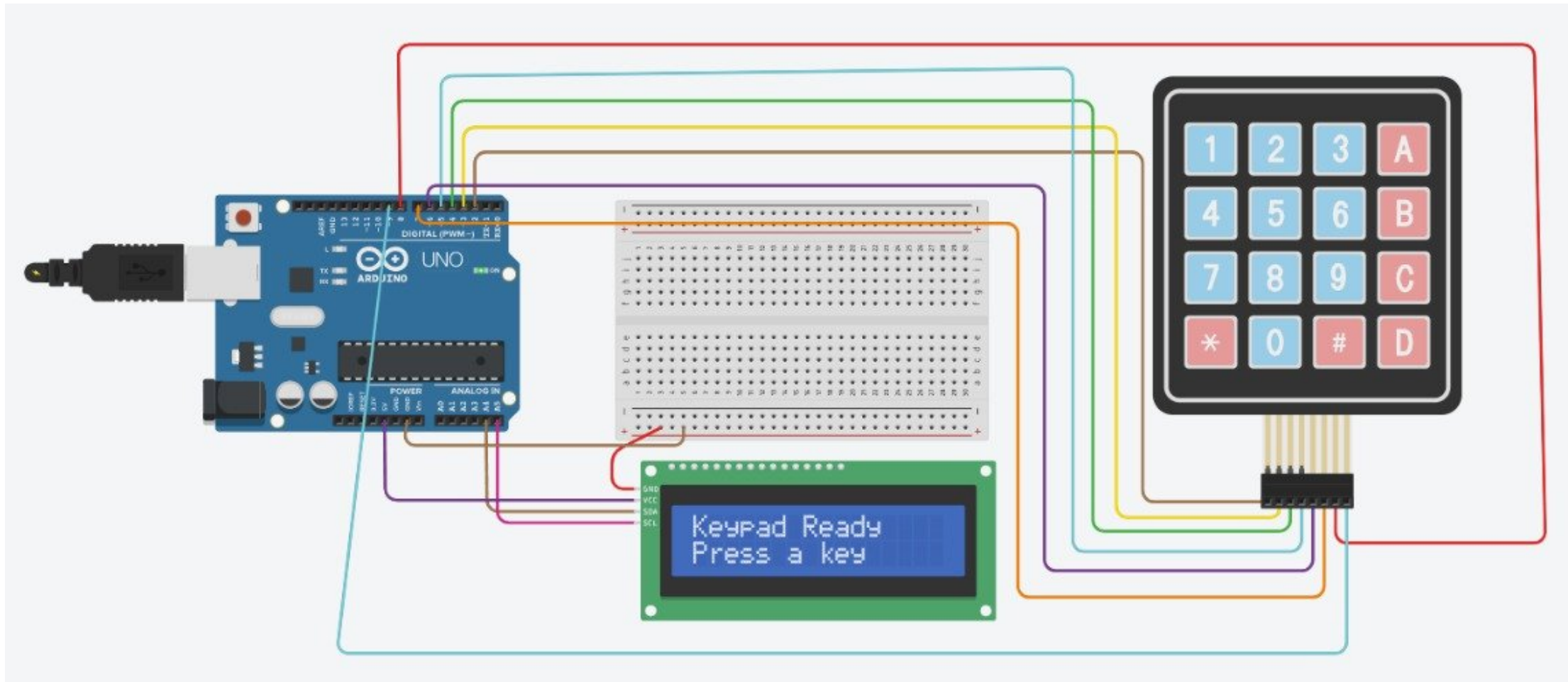


Outputs

- AMOLED display
- Speaker
- Vibration motor
- Cellular communication
- Wi-Fi communication
- Bluetooth communication
- Camera storage/results

This gives a simple explanation of how the phone operates as an engineered system.

TINKCAD SIMULATION



FINAL REVERSE-ENGINEERING CONCLUSIONS

The Samsung Galaxy J7 Duo is a compact smartphone that combines several engineering systems into one portable product. Its major subsystems include the battery and power-management system, processor, memory, display, camera modules, sensors, audio system and wireless communication system.

The external inspection shows that the phone was designed to provide multiple functions while maintaining a relatively compact form factor. Its 5.5-inch Super AMOLED display provides the primary visual interface, while the processor and memory system perform the required computational tasks. The dual rear-camera system provides enhanced photography capabilities, while sensors and the fingerprint system provide additional functionality.

The reverse-engineering process demonstrates that the phone is not simply a collection of individual components. Each component has an electrical and functional purpose, and the components must work together as a complete system.

The major engineering trade-offs include **performance versus power consumption, battery capacity versus size and weight, camera capability versus internal space, and features versus manufacturing cost.**

Therefore, studying the Galaxy J7 Duo through reverse engineering helps reveal the engineering decisions involved in creating a consumer electronic product and demonstrates how mechanical, electrical and software systems are integrated into a single device.

— **THANK YOU**